HARD · ALGEBRA · ANSWER KEY

Algebra

07

10 answers · Rationale-rich · Teach from doc

Q1 D

D. $\frac{5}{7}$

Choice D is correct. It's given that line h is defined by $\frac{1}{5}x + \frac{1}{7}y - 70 = 0$. This equation can be written in slope-intercept form $y = mx + b$, where m is the slope of line h and b is the y-coordinate of the y-intercept of line h. Adding 70 to both sides of $\frac{1}{5}x + \frac{1}{7}y - 70 = 0$ yields $\frac{1}{5}x + \frac{1}{7}y = 70$. Subtracting $\frac{1}{5}x$ from both sides of this equation yields $\frac{1}{7}y = -\frac{1}{5}x + 70$. Multiplying both sides of this equation by 7 yields $y = -\frac{7}{5}x + 490$. Therefore, the slope of line h is $-\frac{7}{5}$. It's given that line j is perpendicular to line h in the xy-plane. Two lines are perpendicular if their slopes are negative reciprocals, meaning that the slope of the first line is equal to -1 divided by the slope of the second line. Therefore, the slope of line j is the negative reciprocal of the slope of line h. The negative reciprocal of $-\frac{7}{5}$ is $\frac{-1}{-\frac{7}{5}}$, or $\frac{5}{7}$. Therefore, the slope of line j is $\frac{5}{7}$.

Choice A is incorrect. This is the slope of a line in the xy-plane that is parallel, not perpendicular, to line h.

Choice B is incorrect. This is the reciprocal, not the negative reciprocal, of $-\frac{7}{5}$.

Choice C is incorrect. This is the negative, not the negative reciprocal, of $-\frac{7}{5}$.

Q2 3331

The correct answer is $3{,}331$. For engine speeds between $1{,}000$ revolutions per minute (rpm) and $6{,}000$ rpm, it's given that the equation $f(x) = \frac{1}{7}(x - a) + 433$ defines the linear function f and gives the predicted power, in brake horsepower (bhp), at an engine speed of x rpm, where a is a constant. It's also given that the car's predicted power is 433 bhp at an engine speed of $3{,}331$ rpm. Substituting $3{,}331$ for x and 433 for $f(x)$ in the equation yields $433 = \frac{1}{7}(3{,}331 - a) + 433$. Subtracting 433 from both sides of this equation yields $0 = \frac{1}{7}(3{,}331 - a)$. Multiplying both sides of this equation by 7 yields $0 = 3{,}331 - a$. Adding a to both sides of this equation yields $a = 3{,}331$. Thus, the value of a is $3{,}331$.

Q3

D

D. -15

Choice D is correct. The line representing the boundary of the shaded region shown is a horizontal line; therefore, the equation of the line can be written as $y = -4$. Since the shaded region represents all of the points above this boundary line, it follows that the shaded region represents the solutions to the inequality $y > -4$. It's given that the shaded region shown represents the solutions to the inequality $ry < 60$, where r is a constant. The inequality $y > -4$ can be rewritten in the form $ry < 60$ by multiplying both sides of the inequality by -15, which yields $(-15)y < 60$. Therefore, the value of r is -15.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Q4

B

B. $\frac{1}{7}$

Choice B is correct. A linear equation in one variable has no solution if and only if the equation is false; that is, when there is no value of x that produces a true statement. It's given that in the equation $-3x + 21px = 84$, p is a constant and the equation has no solution for x. Therefore, the value of the constant p is one that results in a false equation. Factoring out the common factor of $-3x$ on the left-hand side of the given equation yields $-3x1 - 7p = 84$. Dividing both sides of this equation by -3 yields $x1 - 7p = -28$. Dividing both sides of this equation by $1 - 7p$ yields $x = \frac{-28}{1 - 7p}$. This equation is false if and only if $1 - 7p = 0$. Adding $7p$ to both sides of $1 - 7p = 0$ yields $1 = 7p$. Dividing both sides of this equation by 7 yields $\frac{1}{7} = p$. It follows that the equation $x = \frac{-28}{1 - 7p}$ is false if and only if $p = \frac{1}{7}$. Therefore, the given equation has no solution if and only if the value of p is $\frac{1}{7}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q5

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Answer not provided in source data — see rationale below.

The correct answer is $\frac{3}{2}$. One method for solving the system of equations for y is to add corresponding sides of the two equations. Adding the left-hand sides gives $(-x+y)+(x+3y)$, or 4y. Adding the right-hand sides yields $-3.5+9.5=6$. It follows that $4y=6$. Finally, dividing both sides of $4y=6$ by 4 yields $y=\frac{6}{4}$ or $\frac{3}{2}$. Note that 3/2 and 1.5 are examples of ways to enter a correct answer.

Q6

33

The correct answer is 33. It's given in the table that the coordinates of two points on a line in the xy-plane are $(k, 13)$ and $(k+7, -15)$. The y-intercept is another point on the line. The slope computed using any pair of points from the line will be the same. The slope of a line, m, between any two points, x_1, y_1 and x_2, y_2, on the line can be calculated using the slope formula, $m = \frac{y_2 - y_1}{x_2 - x_1}$. It follows that the slope of the line with the given points from the table, $(k, 13)$ and $(k+7, -15)$, is $m = \frac{-15 - 13}{k+7-k}$, which is equivalent to $m = \frac{-28}{7}$, or $m = -4$. It's given that the y-intercept of the line is $(k-5, b)$. Substituting -4 for m and the coordinates of the points $(k-5, b)$ and $(k, 13)$ into the slope formula yields $-4 = \frac{13-b}{k-k-5}$, which is equivalent to $-4 = \frac{13-b}{k-k+5}$, or $-4 = \frac{13-b}{5}$. Multiplying both sides of this equation by 5 yields $-20 = 13 - b$. Subtracting 13 from both sides of this equation yields $-33 = -b$. Dividing both sides of this equation by -1 yields $b = 33$. Therefore, the value of b is 33.

Q7

A

A. $C(d) = 26d + 26$

Choice A is correct. It's given that the cost of renting a carpet cleaner is $\$ 52$ for the first day and $\$ 26$ for each additional day. Therefore, the cost Cd, in dollars, of renting the carpet cleaner for d days is the sum of the cost for the first day, $\$ 52$, and the cost for the additional $d - 1$ days, $\$ 26d - 1$. It follows that $Cd = 52 + 26d - 1$, which is equivalent to $Cd = 52 + 26d - 26$, or $Cd = 26d + 26$.

Choice B is incorrect. This function gives the cost of renting a carpet cleaner for d days if the cost is $\$ 78$, not $\$ 52$, for the first day and $\$ 26$ for each additional day.

Choice C is incorrect. This function gives the cost of renting a carpet cleaner for d days if the cost is $\$ 26$, not $\$ 52$, for the first day and $\$ 52$, not $\$ 26$, for each additional day.

Choice D is incorrect. This function gives the cost of renting a carpet cleaner for d days if the cost is $\$ 130$, not $\$ 52$, for the first day and $\$ 52$, not $\$ 26$, for each additional day.

Q8 D

D. $14, 0$

Choice D is correct. A point x, y is a solution to a system of inequalities in the xy-plane if substituting the x-coordinate and the y-coordinate of the point for x and y, respectively, in each inequality makes both of the inequalities true. Substituting the x-coordinate and the y-coordinate of choice D, 14 and 0, for x and y, respectively, in the first inequality in the given system, $y \leq x + 7$, yields $0 \leq 14 + 7$, or $0 \leq 21$, which is true. Substituting 14 for x and 0 for y in the second inequality in the given system, $y \geq -2x - 1$, yields $0 \geq -214 - 1$, or $0 \geq -29$, which is true. Therefore, the point $14, 0$ is a solution to the given system of inequalities in the xy-plane.

Choice A is incorrect. Substituting -14 for x and 0 for y in the inequality $y \leq x + 7$ yields $0 \leq -14 + 7$, or $0 \leq -7$, which is not true.

Choice B is incorrect. Substituting 0 for x and -14 for y in the inequality $y \geq -2x - 1$ yields $-14 \geq -20 - 1$, or $-14 \geq -1$, which is not true.

Choice C is incorrect. Substituting 0 for x and 14 for y in the inequality $y \leq x + 7$ yields $14 \leq 0 + 7$, or $14 \leq 7$, which is not true.

Q9

B

B. $6n + 6 = 30$

Choice B is correct. It's given that each side of a 30-sided polygon has one of three lengths. It's also given that the number of sides with length 8 centimeters cm is 5 times the number of sides n with length 3 cm. Therefore, there are $5 \times n$, or $5n$, sides with length 8 cm. It's also given that there are 6 sides with length 4 cm. Therefore, the number of 3 cm, 4 cm, and 8 cm sides are n, 6, and $5n$, respectively. Since there are a total of 30 sides, the equation $n + 6 + 5n = 30$ represents this situation. Combining like terms on the left-hand side of this equation yields $6n + 6 = 30$. Therefore, the equation that must be true for the value of n is $6n + 6 = 30$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q10 D

D. $-\frac{6r}{7}+\frac{5}{7}, r$

Choice D is correct. Dividing each side of the second equation in the given system by 4 yields $7x+6y=5$. It follows that the two equations in the given system are equivalent and any point that lies on the graph of one equation will also lie on the graph of the other equation. Substituting r for y in the equation $7x+6y=5$ yields $7x+6r=5$. Subtracting $6r$ from each side of this equation yields $7x=-6r+5$. Dividing each side of this equation by 7 yields $x=-\frac{6r}{7}+\frac{5}{7}$. Therefore, the point $-\frac{6r}{7}+\frac{5}{7}, r$ lies on the graph of each equation in the xy-plane for each real number r.

Choice A is incorrect. Substituting r for x in the equation $7x+6y=5$ yields $7r+6y=5$. Subtracting $7r$ from each side of this equation yields $6y=-7r+5$. Dividing each side of this equation by 6 yields $y=-\frac{7r}{6}+\frac{5}{6}$. Therefore, the point $r, -\frac{7r}{6}+\frac{5}{6}$, not the point $r, -\frac{6r}{7}+\frac{5}{7}$, lies on the graph of each equation.

Choice B is incorrect. Substituting r for x in the equation $7x+6y=5$ yields $7r+6y=5$. Subtracting $7r$ from each side of this equation yields $6y=-7r+5$. Dividing each side of this equation by 6 yields $y=-\frac{7r}{6}+\frac{5}{6}$. Therefore, the point $r, -\frac{7r}{6}+\frac{5}{6}$, not the point $r, \frac{7r}{6}+\frac{5}{6}$, lies on the graph of each equation.

Choice C is incorrect. Substituting $\frac{r}{4}+5$ for x in the equation $7x+6y=5$ yields $7\frac{r}{4}+5+6y=5$, or $\frac{7r}{4}+35+6y=5$. Subtracting $\frac{7r}{4}+35$ from each side of this equation yields $6y=-\frac{7r}{4}-35+5$, or $6y=-\frac{7r}{4}-30$. Dividing each side of this equation by 6 yields $y=-\frac{7r}{24}-5$. Therefore, the point $\frac{r}{4}+5, -\frac{7r}{24}-5$, not the point $\frac{r}{4}+5, -\frac{r}{4}+20$, lies on the graph of each equation.

Notes

Accept equivalent forms (e.g. `0.25 ≡ 1/4`). For grid-in items, decimal and fraction answers are both valid.

Answers recovered from rationale for Q5 — the source CB data omitted a structured key for these items, so the answer slot shows –. The worked solution in the rationale below each entry contains the correct value.

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